Digital Game Design 🎮

Junior Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

Games are built *for players* 💚

A good designer asks: - **Who** plays this? - Is it **fair**? - Is it **fun for everyone**?

That's **positive gaming** — the "for good" of this badge.

Dissect a game like a designer 🔍

Pick a game everyone knows. Break it into its designer parts:

Part	The question
Goal	what are you trying to do?
Rules	what are you allowed to do?
Win condition	how does it end?
Difficulty curve	does it get harder as you go?

Write the design doc 

Every real studio starts here. One page:

1. **Goal**
2. **Rules**
3. **Win condition**
4. What makes it **fun**
5. **Who it's for**

Scope rule: one goal, one obstacle, one way to win.

No screens until the doc is done ✋

It’s the difference between **designing** and **doodling**.

Build YOUR game 💻

In MakeCode Arcade — building *your* design doc, not a clone:

1. Player sprite with arrow-key movement
2. Collectible + a **score counter**
3. **Win condition** — hit the target score → "YOU WIN"
4. **Challenge element** — an enemy that chases, or a countdown timer
5. Stretch: a second level, or increasing speed — that's your **difficulty curve**

Structured Playtest 🔄

Swap seats. Play for **two minutes**. Leave two sticky notes:

- ⭐ **One star** — what's fun
- 💡 **One wish** — one change that would make it better

Always both. Feedback is a gift.

Now iterate 🔧

Make **ONE improvement** based on your feedback.

Iteration is the most important habit in game design.

Robot moment 

A brand-new competition game drops **every season**.

Our team's very first job is exactly this analysis: goals, rules, win conditions, strategy.

Closing question 🙋

Who changed their game after feedback?

(Everyone should.)

What you accomplished ✅

- Broke a game into goal, rules, win condition, difficulty
- Wrote a one-page design doc before building
- Built a playable game with a win condition and a challenge
- Ran a playtest and improved your game from real feedback

Keep going at home! 🏠

- **arcade.makecode.com** — finish your stretch goals (free)
- Write a design doc for your dream game
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.